

STEP Support Programme — Assignment 1

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We will only write up problems 3 and 4, since the others are trivial.

Problem 3

In this question a and b are distinct, non-zero real numbers, and c is a real number.

- (i) Show that, if a and b are either both positive or both negative, then the equation

$$\frac{x}{x-a} + \frac{x}{x-b} = 1$$

has two distinct real solutions.

- (ii) Show that, if $c \neq 1$, the equation

$$\frac{x}{x-a} + \frac{x}{x-b} = 1 + c$$

has exactly one real solution if

$$c^2 = -\frac{4ab}{(a-b)^2}.$$

Show that this condition can be written

$$c^2 = 1 - \left(\frac{a+b}{a-b}\right)^2$$

and deduce that it can only hold if $0 < c^2 \leq 1$.

Solution to Problem 3

For part (i), we have $2x^2 - (a+b)x = x^2 - (a+b)x + ab$ and so $x^2 = ab$. If a and b have the same parity then $ab > 0$ (since a, b are non-zero) and so $x^2 = ab$ will have 2 distinct real solutions, namely $\sqrt{ab}$ and $-\sqrt{ab}$.

In part (ii) we get $2x^2 - (a+b)x = (1+c)x^2 - (a+b)x - c(a+b)x + ab + abc$ and so $(c-1)x^2 - c(a+b)x + ab(c+1) = 0$. To have exactly one real solution the discriminant must be 0 so, $[c(a+b)]^2 = 4ab(c-1)(c+1)$. Now observe that,

$$\begin{aligned} c^2 &= \frac{4(c^2-1)ab}{(a+b)^2} \\ &= \frac{4c^2}{(a+b)^2} - \frac{4ab}{(a+b)^2} \\ c^2 \left(1 - \frac{4ab}{(a+b)^2}\right) &= -\frac{4ab}{(a+b)^2} \\ c^2 \frac{(a-b)^2}{(a+b)^2} &= -\frac{4ab}{(a+b)^2} \end{aligned}$$

and so we readily derive $c^2 = -\frac{4ab}{(a-b)^2}$, as desired. To do the next part we can work backwards, we have

$$\begin{aligned} c^2 &= 1 - \left(\frac{a+b}{a-b} \right)^2 \iff \\ &= \left(1 - \frac{a+b}{a-b} \right) \left(1 + \frac{a+b}{a-b} \right) \iff \\ &= \left(-\frac{2b}{a-b} \right) \left(\frac{2a}{a-b} \right) \iff \\ &= -\frac{4ab}{(a-b)^2} \end{aligned}$$

but that is obviously true as we just showed that before. Now finally to show that inequality, we know that for any real number x , we have $x^2 \geq 0$ and so $c^2 \geq 0$. But $c^2 \neq 0$ since if $c^2 = -\frac{4ab}{(a-b)^2} = 0$ then one of a, b is 0, but this is impossible (*In this question a and b are distinct, non-zero real numbers...*). Thus we have established $c^2 > 0$. To show that $c^2 \leq 1$, note that $\left(\frac{a+b}{a-b} \right)^2 \geq 0$ and so $c^2 = 1 - \left(\frac{a+b}{a-b} \right)^2 \leq 1$, as desired.

Problem 4

The diagram shows a circle C with centre O , and a rod AB the ends of which can slide round the circle C (so that AB is a chord of C). The radius of the circle is R and the length of the rod is $2a$. As the rod slides round C the point P , which is a fixed distance b from the centre of the rod, traces out a circle with centre O of radius r .

Show that the area between the 2 circles is $\pi(a^2 - b^2)$.

Solution to Problem 4

Observe the following diagram

